

RECAP: 1) Random vectors:

- $X = [X_1, X_2, \dots, X_m]^T$ of dimension m , X_i are RVs defined on $(\Omega, \mathcal{F}, P)$
- Mean $E[X] = [E[X_1], E[X_2], \dots, E[X_m]]^T$
- Autocorrelation: $E[XX^T]$ - $m \times m$ matrix with $(i, j)^{th}$ entry $E[X_i X_j]$.
- Autocovariance: $cov(X)$ - $m \times m$ matrix with $(i, j)^{th}$ entry $cov(X_i, X_j)$.
- Cross correlation $E[XY^T]$ - for two random vectors X & Y on same $(\Omega, \mathcal{F}, P)$
- Cross-covariance $cov(X, Y)$

2) Random Processes.

- Definition: $\{X_t, t \in T\}$, collection of RVs on the same $(\Omega, \mathcal{F}, P)$
- T : index set; if T countably infinite: discrete parameter process
if T uncountably infinite: continuous parameter process
- Three ways to look at a Random Process:
 - for a fixed $t \in T$, $X_t: \Omega \rightarrow \mathbb{R}$ a random variable
 - for a fixed $\omega \in \Omega$, $X(\omega): T \rightarrow \mathbb{R}$, a sample path
 - as a function of two variables $X(t, \omega): T \times \Omega \rightarrow \mathbb{R}$
- Finite dimensional distribution: $F_{t_1, t_2, \dots, t_m}(x_1, x_2, \dots, x_m)$ - Consistency.
- Properties: Mean: $\mu_X(t)$, Autocorrelation: $R_X(t, s)$, Autocovariance: $C_X(t, s)$

PLAN:

- 1) Examples of Random processes: Gambler's Ruin.
- 2) Stationary Processes: Strict Sense, N^{th} Order, Wide Sense.

References: 1) Bruce Hajek, "Random Processes for Engineers"

2) Robert G. Gallager, "Stochastic Processes: Theory for Applications"

Homework from 01 August

Example 2 : $X_1 \sim \text{unif}[0,1]$, $X_2 = X_1 + W$, $W \sim \text{unif}[0,1]$ $X_1 \perp W$

Let $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$. Find $\mathbb{E}[X]$, $\mathbb{E}[XX^T]$ and $\text{cov}(X)$. (Homework)

$$\mathbb{E}[X] = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}, \quad \mathbb{E}[XX^T] = \begin{bmatrix} 1/3 & 7/12 \\ 7/12 & 7/6 \end{bmatrix}, \quad \text{cov}(X) = \begin{bmatrix} 1/12 & 1/12 \\ 1/12 & 1/6 \end{bmatrix}$$

Example 3 : Consider $X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$, where X_1, X_2 are random variables defined on the same probability space with joint distribution

$$f_{X_1, X_2}(x_1, x_2) = \begin{cases} 6x_2, & 0 \leq x_1 \leq 1, 0 \leq x_2 \leq x_1 \\ 0 & \text{otherwise} \end{cases}$$

Find $\mathbb{E}[X]$ and $\mathbb{E}[XX^T]$

$$\mathbb{E}[X] = \begin{bmatrix} 3/4 \\ 1/2 \end{bmatrix}$$

$$\mathbb{E}[XX^T] = \begin{bmatrix} 3/5 & 2/5 \\ 2/5 & 3/10 \end{bmatrix}$$

(Homework)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Defn: A second order random process is a random process $\{X_t : t \in \mathcal{T}\}$ such that $\mathbb{E}[X_t^2] < \infty \forall t \in \mathcal{T}$.

The mean, correlation and covariance functions of a second order random process are all well-defined and finite.

Consider a random process $\{X_t : t \in \mathcal{T}\}$ defined over a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$

- If X_t is a discrete random variable for each $t \in \mathcal{T}$, then the n^{th} order PMF of the RP

$\{X_t : t \in \mathcal{T}\}$ is defined by $P_{t_1, t_2, \dots, t_n}(x_1, \dots, x_n) = \mathbb{P}(\{X_{t_1} = x_1, \dots, X_{t_n} = x_n\})$

- Similarly if $X_{t_1}, X_{t_2}, \dots, X_{t_n}$ are jointly continuous random variables for any distinct $t_1, t_2, \dots, t_n \in \mathcal{T}$, then $\{X_t : t \in \mathcal{T}\}$ has an n^{th} order pdf s.t for each $(t_1, t_2, \dots, t_n)$ fixed, $f_{t_1, \dots, t_n}(x_1, \dots, x_n)$ is the joint pdf of $X_{t_1}, X_{t_2}, \dots, X_{t_n}$.

Example: Let A and B be iid $N(0,1)$ random variables.

Suppose $X_t = A + Bt + t^2 \forall t \in \mathbb{R}$.

Describe $\mu_X(t)$, $R_X(s,t)$, $C_X(s,t)$, the first and second order pdfs on $\{X_t : t \in \mathbb{R}\}$

$$\mu_X(t) = \mathbb{E}[X_t] = \mathbb{E}[A + Bt + t^2] = t^2$$

$$R_X(s,t) = \mathbb{E}[X_s X_t] = \mathbb{E}[(A + Bs + s^2)(A + Bt + t^2)] = 1 + st + s^2 t^2$$

$$C_X(s,t) = R_X(s,t) - \mu_X(s)\mu_X(t) = 1 + st + s^2 t^2 - s^2 t^2 = 1 + st$$

For a fixed t , X_t is a linear combination of 2 $N(0,1)$ RVs $\therefore$ a Gaussian RV with mean $\mu_X(t) = t^2$ and variance $C_X(t,t) = 1 + t^2 \therefore X_t \sim N(t^2, 1+t^2)$

$\therefore$ First order pdf: $f_{t_1}(x_1) = \frac{1}{\sqrt{2\pi(1+t^2)}} \exp\left(-\frac{(x-t^2)^2}{2(1+t^2)}\right)$ $X_s = A + Bs + s^2$
 $X_t = A + Bt + t^2$

For distinct s and t , $\text{Cov}(X_s, X_t) = \begin{bmatrix} 1+s^2 & 1+st \\ 1+st & 1+t^2 \end{bmatrix}$ $|\text{Cov}(X_s, X_t)| = (s-t)^2 \neq 0$

$\therefore \{X_t, t \in \mathbb{R}\}$ has a second order pdf.
 $f_{s,t} = \frac{1}{2\pi|s-t|} \exp\left\{-\frac{1}{2} \begin{pmatrix} x-s^2 \\ y-t^2 \end{pmatrix}^T \begin{bmatrix} 1+s^2 & 1+st \\ 1+st & 1+t^2 \end{bmatrix}^{-1} \begin{pmatrix} x-s^2 \\ y-t^2 \end{pmatrix}\right\}$

Example: IID Process.

Consider a prob. space $(\Omega, \mathcal{F}, P)$

$X_1, X_2, X_3, \dots$ defined w.r.t $\mathcal{F}$, i.i.d with $P(X_i = 1) = p$ and $P(X_i = -1) = 1-p$

$\{X_i; i \in \mathbb{N}\}$ is an i.i.d Bernoulli process.

$F_{t_1, t_2, \dots, t_n}(x_1, \dots, x_n) = \prod_{i=1}^n F_{t_i}(x_i)$ Are they consistent?

$\mu_X(t) = 2p-1$

$R_X(t, s) = (2p-1)^2$

$C_X(t, s) = 0$

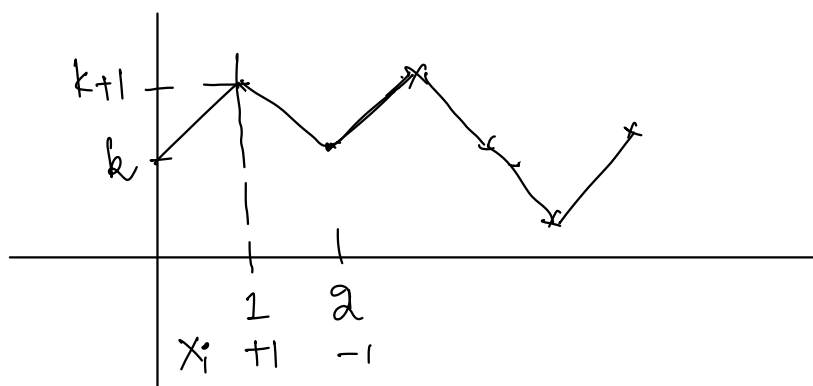
$\{X_i\}_{i \in \mathbb{N}} \quad X_i \stackrel{\text{i.i.d}}{\sim} \text{Ber}(p)$

Let X_0 be an integer valued random variable on the same prob. space independent of $X_1, X_2, \dots$

We define $Z_n = \sum_{i=0}^n X_i$ $\{Z_n: n \geq 0\}$ is a random walk

What is $E[Z_n | X_0 = k]$? $\text{Var}(Z_n | X_0 = k) = ?$ $Z_n = X_0 + X_1 + \dots + X_n$

$E[Z_n | X_0 = k] = k + n(2p-1)$
 $\text{Var}(Z_n | X_0 = k) = n(2p-1)^2$



Gambler's Ruin: Gambler's initial wealth is k ; $k \geq 0$

Z_n is the number of units of money he has at time n .

Let the gambler has a goal of accumulating b units of money, $b > k$. While the random walk $\{Z_n: n \geq 0\}$ continues forever, we are only interested in it until it hits either 0 (gambler is ruined) or b (the gambler is successful).

Let S_b denote the event that the gambler is successful, meaning the random walk reaches b before reaching 0.

Gambler's success prob. $s_k \triangleq P(S_b | X_0 = k)$ — success probability for a gambler with initial wealth k and target wealth b .

$s_0 = 0, s_b = 1$ (Boundary Conditions)

$$s_k = P\{X_1 = 1\} P\{S_b | X_1 = 1, X_0 = k\} + P\{X_1 = -1\} P\{S_b | X_1 = -1, X_0 = k\}$$
$$= p s_{k+1} + (1-p) s_{k-1} \quad \text{for } 1 \leq k \leq b-1$$

This yields $(b-1)$ linear equations in $(b-1)$ unknowns $s_1, \dots, s_{b-1}$

When $p = 1/2$, $s_k = \frac{1}{2} \{s_{k+1} + s_{k-1}\}$, $1 \leq k \leq b-1$

With boundary conditions $s_0 = 0$ and $s_b = 1$, we get $s_k = k/b$.

Note that when the gambler stops playing, he has b units of money with prob. k/b and 0 units otherwise $\therefore$ His expected wealth is $b \cdot \frac{k}{b} + 0 \cdot (1 - \frac{k}{b}) = \underline{k}$ (his initial wealth)

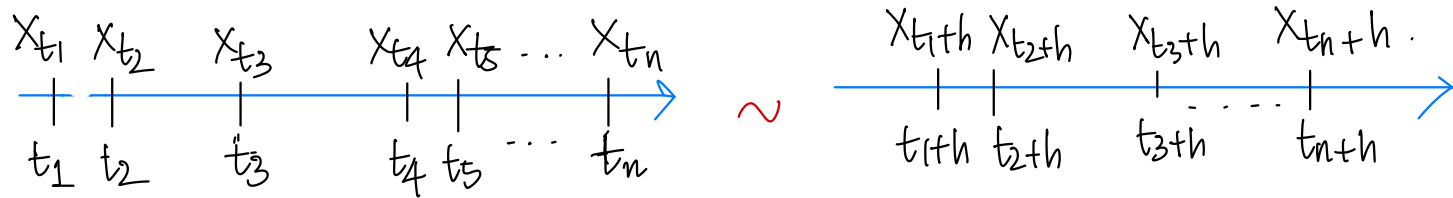
Stationary Process:

Fix a probability space $(\Omega, \mathcal{F}, P)$

Let $\{X_t, t \in \mathbb{R}\}$ be a random process defined w.r.t $\mathcal{F}$

$$F_{t_1}(x_1) = F_{t_2}(x_1) \quad \forall t_1, t_2$$
$$F_{t_1, t_2}(x_1, x_2) = F_{t_1+h, t_2+h}(x_1, x_2)$$
$$\forall t_1, t_2, h \in \mathbb{R}$$

Defn: The random process $\{X_t, t \in \mathbb{R}\}$ is said to be **(strictly) stationary** if all FDDs are **translation-invariant**, i.e., for any $n \in \mathbb{N}$, $(t_1, \dots, t_n) \in \mathbb{R}^n$ and $h \in \mathbb{R}$, $F_{t_1, \dots, t_n}(x_1, \dots, x_n) = F_{t_1+h, \dots, t_n+h}(x_1, \dots, x_n)$, $\forall (x_1, \dots, x_n) \in \mathbb{R}^n$



Nth Order Stationary Process:

A random process $\{X_t, t \in \mathbb{R}\}$ defined w.r.t the prob. space $(\Omega, \mathcal{F}, P)$ is said to be

— **first-order stationary** if $F_t(x) = F_{t+h}(x) \quad \forall t, h \in \mathbb{R}, x \in \mathbb{R}$.

— **second-order stationary** if $F_{t_1, t_2}(x_1, x_2) = F_{t_1+h, t_2+h}(x_1, x_2) \quad \forall t_1, t_2, h \in \mathbb{R}, x_1, x_2 \in \mathbb{R}$
and $F_t(x) = F_{t+h}(x), \quad \forall t, h \in \mathbb{R}, x \in \mathbb{R}$

In particular, if we take $h = -t_1$, we have $F_{t_1, t_2}(x_1, x_2) = F_{0, t_2-t_1}(x_1, x_2)$

The second order joint distribution only depends on the shift $t_2 - t_1$ and not on the values of t_2 and t_1 .

$R_X(t_1, t_2) = E[X_{t_1} X_{t_2}]$ will be a function of only $t_2 - t_1$ and not the actual values of t_2 and t_1 . $\therefore$ For a 2nd order stationary process, the autocorrelation

fn is denoted as $R_X(t_2 - t_1)$ or $R_X(\Delta)$ where Δ denotes the shift, $\Delta \in \mathbb{R}$

— **Nth order stationary** if $F_{t_1, \dots, t_n}(x_1, \dots, x_n) = F_{t_1+h, \dots, t_n+h}(x_1, \dots, x_n)$, for all $t_1, \dots, t_n \in \mathbb{R}$ and $x_1, \dots, x_n \in \mathbb{R}, \quad \forall n \in \{1, 2, \dots, N\}$.

Wide Sense Stationary (WSS) Process

Fix a probability space $(\Omega, \mathcal{F}, P)$

Let $\{X_t : t \in \mathbb{R}^+\}$ be a random process defined w.r.t $\mathcal{F}$.

Defn: $\{X_t : t \in \mathbb{R}\}$ is said to be **wide-sense stationary** (OR **weakly stationary**)

if for all $t_1, t_2 \in \mathbb{R}^+$ and $h \in \mathbb{R}^+$

1) $\mu_X(t_1) = \mu_X(t_2) \Rightarrow \mu_X$ is a constant

2) $C_X(t_1, t_2) = C_X(t_1+h, t_2+h) \Rightarrow \underbrace{C_X(t_2-t_1)}_{R_X(t_2-t_1)}$

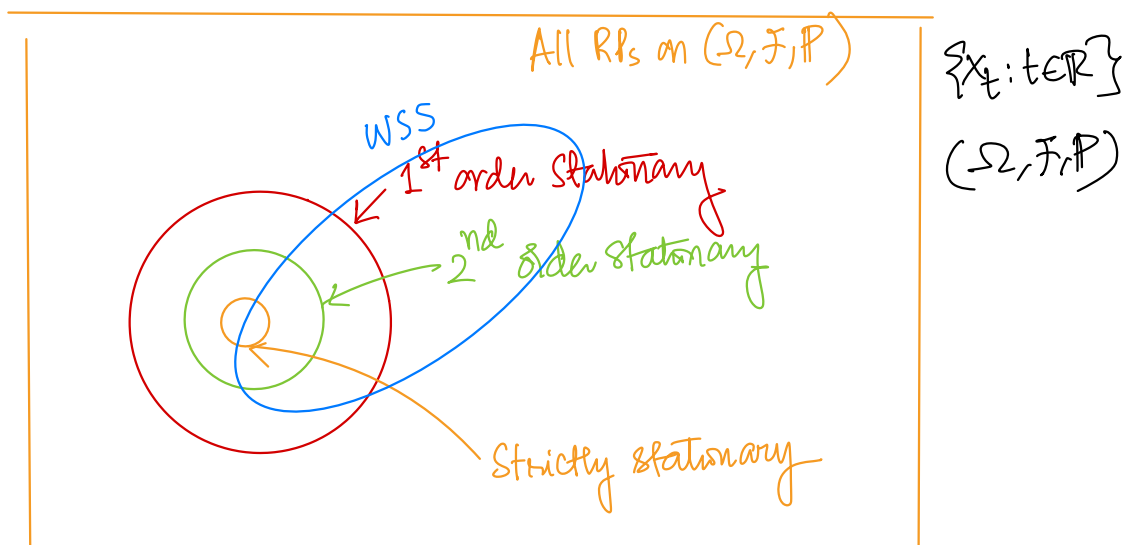
3) $E[X_t^2] < \infty, \forall t \in \mathbb{R}$

Prove 1: Every stationary process with finite variance is WSS.

2) A process $\{X_t, t \in \mathbb{R}\}$ is WSS iff it has constant mean & $C_X(t, t+h) = \underline{C_X(0, h)}$

HOMEWORK!!

Qn: Where will WSS processes lie? (Containment relation)



Example: 1) Stationary but not WSS. : IID Cauchy Process.

Fix a prob. space $(\Omega, \mathcal{F}, \mathbb{P})$. Consider $\{X_t : t \in \mathcal{T}\}$ where $X_t \text{ iid Cauchy}(0,1) \forall t \in \mathcal{T}$

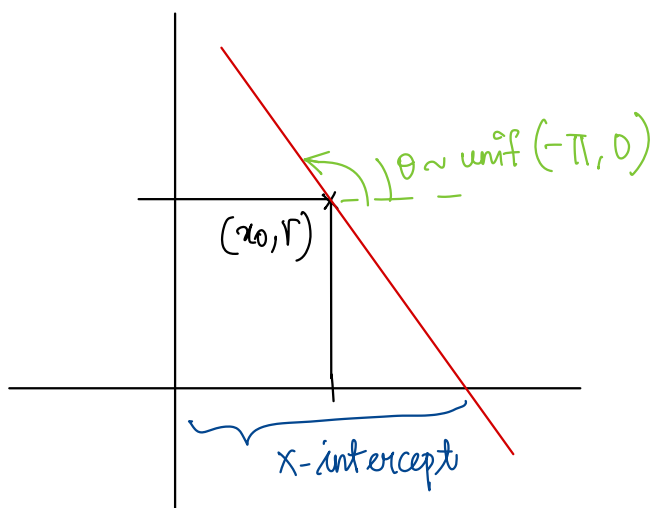
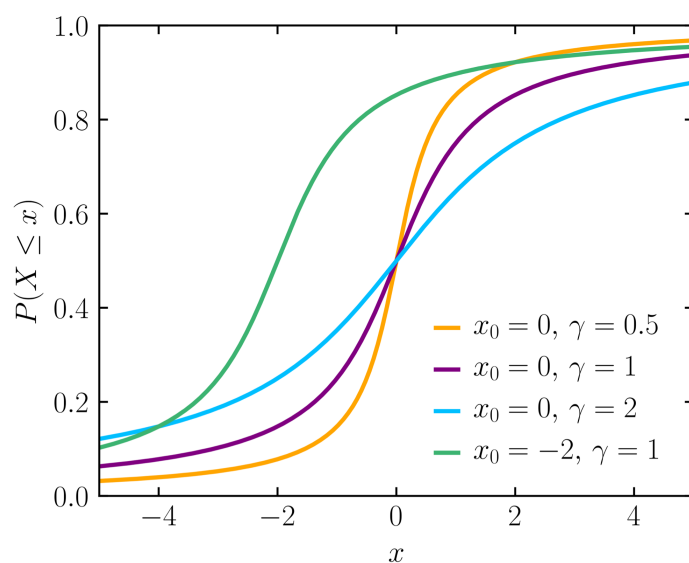
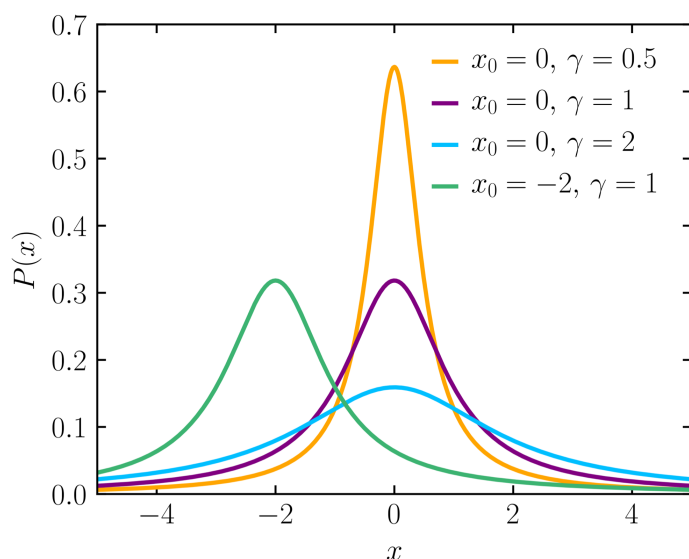
Cauchy (Lorentzian) Distribution :

A continuous RV. $X \sim \text{Cauchy}(0,1)$ has a pdf : $f_X(x) = \frac{1}{\pi(1+x^2)}$, $x \in (-\infty, \infty)$

In general: if $X \sim \text{Cauchy}(x_0, r)$, $f_X(x) = \frac{1}{\pi r \left[1 + \frac{(x-x_0)^2}{r^2} \right]}$, $x \in (-\infty, \infty)$

$$\text{CDF: } F_X(x) = \frac{1}{\pi} \tan^{-1} \left(\frac{x-x_0}{r} \right) + \frac{1}{2}$$

x_0 : Location
 r : Scale.



$f_X(x, x_0, r)$ is the distribution of the x -intercept of a line passing through (x_0, r) with a uniformly distributed angle.

Generating a sample from a Cauchy $(0,1)$ distribution:

Let u be a sample from $\text{unif}(0,1)$.

$x = \tan(\pi(u - 1/2))$ is a sample from $\text{Cauchy}(0,1)$

Cauchy distribution has no mean, variance or higher moments defined..

$\therefore$ IID Cauchy process is stationary but not WSS!!

Example: 2) WSS but not stationary

$$X(t) = A \cos(2\pi f t + \Theta), \quad t \in \mathbb{R}$$

A, f - constants, Θ is a random variable $\sim \text{unif}[0, 2\pi]$

Mean: $\mu_X(t) = A \int_0^{2\pi} \cos(2\pi f t + \theta) \frac{1}{2\pi} d\theta = 0$

Mean is constant!

Autocorrelation: $R_X(s, t) = \mathbb{E}[X(s)X(t)]$

$$\mathbb{E}[A^2 \cos(2\pi f s + \Theta) \cos(2\pi f t + \Theta)] = \frac{A^2}{2} [\underbrace{\cos(2\pi f (s+t+2\Theta))}_{\text{and finite}} + \underbrace{\cos(2\pi f (s-t))}_{\substack{\text{uncorrelated for} \\ \text{certain pairs } (s, t)}}]$$
$$\mathbb{E}[X(s)X(t)] = \underbrace{\frac{A^2}{2} \cos(2\pi f (s-t))}_{\text{fn of only } (s-t)}$$

$\therefore \{X(t), t \in \mathbb{R}\}$ is a WSS process.

Is it stationary?

$$F_f(x) = \mathbb{P}(A \cos(2\pi f t + \Theta) \leq x)$$